phonons d-dimensional lattice

$$H = \sum_r \frac{1}{2m} \hat{p}_r^\alpha \hat{p}_r^\alpha + \frac{1}{2} \sum_{rr'} (u_r^\alpha - u_{r'}^\alpha)\, \phi_{r-r'}^{\alpha\beta} (u_r^\beta - u_{r'}^\beta)$$

It is possible to diagonalize it by

$$u_r^\alpha = \frac{1}{\sqrt{N}} \sum_{\vec{k}\in BZ} e^{i\vec{k}\cdot\vec{r}}\, T_k^{\alpha\beta}\, u_k^\beta$$

$\downarrow$ $d\times d$ matrix

$$H_{ph} = \sum_{\vec{k}\in BZ} \left(\frac{1}{2m} p_k^\nu p_{-k}^\nu + \frac{1}{2} m \Omega_{k\nu}^2 u_k^\nu u_{-k}^\nu \right)$$

$\uparrow$ phonon frequency of ν-th branch ($\nu = 1, 2 \ldots d$)

Quantize lattice oscillations

$$a_k^\nu = \sqrt{\frac{m\Omega_{k\nu}}{2\hbar}}\, u_k^\nu + \frac{i}{\sqrt{2\hbar m \Omega_{k\nu}}} p_k^\nu$$

$$(a_k^\nu)^+ = \sqrt{\frac{m\Omega_{k\nu}}{2\hbar}}\, \underbrace{u_{-k}^\nu}_{(u_k^\nu)^+} - \frac{i}{\sqrt{2\hbar m \Omega_{k\nu}}} p_{-k}^\nu$$

$$H_{ph} = \sum_{k\in BZ} \hbar\Omega_{k\nu} \left[(a_k^\nu)^+ a_k^\nu + \frac{1}{2} \right]$$

$$\hat{u}_r = \sum_{\substack{k\in BZ \\ \nu}} \sqrt{\frac{\hbar}{2m\Omega_{k\nu} N}} \left(\vec{e}_{k\nu}\, a_{k\nu}\, e^{i\vec{k}\cdot\vec{r}} + \vec{e}_{k\nu}^{\,*}\, a_{-k\nu}^+\, e^{-i\vec{k}\cdot\vec{r}} \right)$$

$\uparrow$ polarization vector

Electron-phonon interaction

Lattice deformation causes polarization charge $\sim \vec{\nabla}\cdot\vec{P}$ ← electric polarization

Electron interacts with it.

Potential energy $\sim e\int f(\vec{r}-\vec{r}\,')\,\vec{\nabla}\cdot\vec{P}(\vec{r}\,')\,dV'$

↑ the kernel is screened; finite on atomic scales only

$\Rightarrow f(r-r') \approx a^2\,\delta(r-r')$

$$\vec{\nabla}\cdot\vec{P} = ne\,\vec{\nabla}\cdot\vec{u} = ne\sum_{k\in BZ}\frac{\hbar\,\vec{k}\cdot}{\sqrt{2m\Omega_{k\nu}N}}\left(i\vec{e}_{k\nu}a_{k\nu}e^{ikr} + i\vec{e}^{\,*}_{k\nu}a^{+}_{-k\nu}e^{-ikr}\right)$$

↑ displacement

$$\hat{H}_{e\text{-}ph} = \int d\vec{r}\,\rho_e(\vec{r})\,\vec{\nabla}\cdot\vec{P}$$

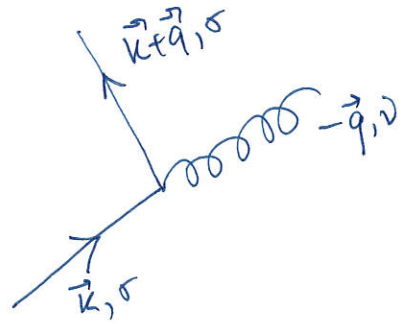
electron density $\rho_e(\vec{r}) = \frac{1}{N}\sum_{kp\sigma} e^{-i\vec{p}\cdot\vec{r}}\,c^{+}_{k+p\sigma}c_{k\sigma}$

$$\hat{H}_{e\text{-}ph} = \frac{1}{N}\sum_{k\sigma}\sum_{q\nu} g_{q,\nu}\,c^{+}_{k+q,\sigma}c_{k\sigma}\left(a_{q\nu}+a^{+}_{-q\nu}\right)$$

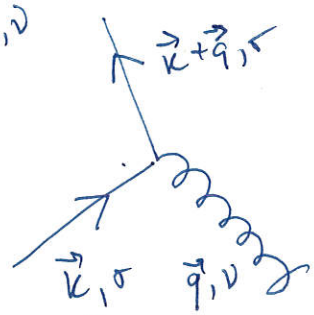
$$g_{q,\nu} = ie\sqrt{\frac{\hbar}{2m\Omega_{q\nu}}}\;\vec{q}\cdot\vec{e}_{q\lambda}$$

matrix element of e-ph coupling

depends on phonon's polarization (strongest for longitudinal phonons)

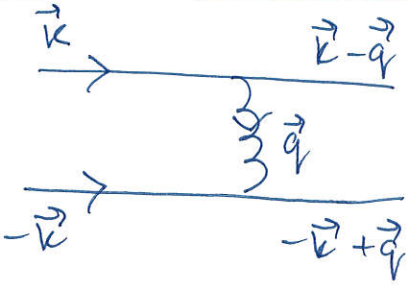


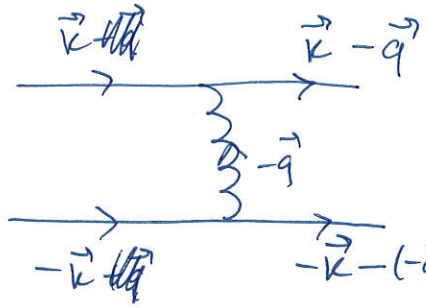
emission of phonon



absorption of phonon

Nature of attraction





$\vec{k}$ emits $\vec{q}$
$-\vec{k}$ absorbs it.

$-\vec{k}+\vec{q}$ emits $-\vec{q}$;
$\vec{k}-\vec{q}$ absorbs

$$\frac{|g_{q,\nu}|^2}{\varepsilon_k - (\varepsilon_{k-q} + \hbar\omega_q)} \qquad \frac{|g_{q,\nu}|^2}{\varepsilon_{-\vec{k}} - (\varepsilon_{-\vec{k}+\vec{q}} + \hbar\omega_{-q})}$$

$$\left(2^{nd} \text{ order pert. theory } \delta E = \frac{|\text{matrix element}|^2}{E_{in} - E_{fi}}\right)$$

Energy conservation $\quad \varepsilon_k + \varepsilon_{-k} = \varepsilon_{k-q} + \varepsilon_{-k+q}$

$$\Rightarrow \varepsilon_k - \varepsilon_{k-q} = \varepsilon_{-k+q} - \varepsilon_{-k}$$

$\Rightarrow$ the sum of two identical processes is:

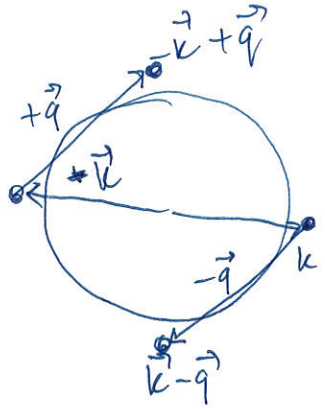
$$|g_{q,\nu}|^2 \left(\frac{1}{\varepsilon_k - \varepsilon_{k-q} - \hbar\omega_q} + \frac{1}{\varepsilon_{-k} - \varepsilon_{-k+q} - \hbar\omega_{-q}} \right)$$

$$= |g_{q,\nu}|^2 \left(\frac{1}{\varepsilon_k - \varepsilon_{k-q} - \hbar\omega_q} + \frac{1}{-(\varepsilon_k - \varepsilon_{k-q}) - \hbar\omega_q} \right)$$

$$= |g_q|^2 \frac{\varepsilon_k - \varepsilon_{k-q} + \hbar\omega_q - (\varepsilon_k - \varepsilon_{k-q} - \hbar\omega_q)}{(\varepsilon_k - \varepsilon_{k-q})^2 - (\hbar\omega_q)^2} =$$

$$= 2|g_q|^2 \frac{\hbar\omega_q}{(\varepsilon_k - \varepsilon_{k-q})^2 - (\hbar\omega_q)^2} \approx -\frac{2|g_q|^2}{\hbar\omega_q}$$

for electrons on the Fermi surface when $\varepsilon_{\vec{k}} = \varepsilon_F$



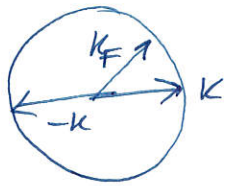
$\Rightarrow$ multiple scattering of electrons by phonons in the vicinity of Fermi surface leads to effective attraction for $|\varepsilon_k - \varepsilon_{k-q}| < \hbar\omega_q$,

i.e. within narrow shell of ~~energy~~ energies centered at E_F.

The width of that shell is of the order of Debye energy ~ few hundred K.

<u>Cooper problem</u> : two electrons at the Fermi surface.
All $k < k_F$ are filled.

L. Cooper
Phys. Rev. 104, 1189 (1958)



Two-body problem for $\psi(\vec{r}_1, \vec{r}_2)$:

$$-\frac{\hbar^2}{2m}\left(\vec{\nabla}_1^2 + \vec{\nabla}_2^2\right)\psi(r_1, r_2) + V(\bar{r}_1 - \bar{r}_2)\,\psi(r_1; r_2) =$$

$$= \underbrace{\left(\varepsilon + 2\cdot\frac{\hbar^2 k_F^2}{2m}\right)}\psi(r_1, r_2)$$

ε, energy of two particles calculated from E_F

Center of mass $\vec{R} = \frac{1}{2}(\bar{r}_1 + \bar{r}_2)$
relative $\vec{r} = \bar{r}_1 - \bar{r}_2$

$$\Rightarrow \quad -\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_2^2\right) = -\frac{\hbar^2}{2M}\nabla_R^2 - \frac{\hbar^2}{2\mu}\nabla_r^2$$

$M = m_1 + m_2 = 2m$ (total mass)

$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m^2}{2m} = \frac{m}{2}$ (reduced mass)

$$-\frac{\hbar^2}{4m}\nabla_R^2\psi - \frac{\hbar^2}{m}\nabla_r^2\psi + V(\vec{r})\psi = \left(\varepsilon + \frac{\hbar^2 k_F^2}{m}\right)\psi$$

assume CM at rest (electrons with momenta $\vec{k}, -\vec{k}$)

$\Rightarrow \psi(\bar{r}_1, \bar{r}_2) = \psi(\vec{r})$ only

Fourier $g(\vec{k}) = \int e^{-i\vec{k}\cdot\vec{r}} \psi(\vec{r})\, d^3\vec{r}$

$$-\frac{\hbar^2}{m}\int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, \nabla_r^2 \psi(r) + \int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, V(r)\psi(r)$$

$$= \left(\varepsilon + \frac{\hbar^2 k_F^2}{m}\right) \underbrace{\int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, \psi(r)}_{g(\vec{k})}$$

surface term $\to 0$

$$\int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, \nabla^2\psi(r) = \int d\vec{r}\, \vec{\nabla}\left(e^{-i\vec{k}\cdot\vec{r}}\vec{\nabla}\psi\right) - \vec{\nabla}\left(e^{-i\vec{k}\cdot\vec{r}}\right)\vec{\nabla}\psi$$

$$= i\vec{k}\int d\vec{r}\, \vec{\nabla}\left(e^{-i\vec{k}\cdot\vec{r}}\psi\right) - \vec{\nabla}\left(e^{-i\vec{k}\cdot\vec{r}}\right)\psi$$

$$= (i\vec{k})^2 \int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, \psi(r) = -k^2 g(\vec{k})$$

$$\int d\vec{r}\, e^{-i\vec{k}\cdot\vec{r}}\, V(r)\psi(r) = \int d\vec{r} \int \frac{d\vec{q}}{(2\pi)^3}\frac{d\vec{k}'}{(2\pi)^3}\, e^{-i\vec{k}\cdot\vec{r}} e^{i\vec{q}\cdot\vec{r}} e^{i\vec{k}'\cdot\vec{r}}$$

$$\tilde{V}(\vec{q})\, g(\vec{k}') = \int \frac{d\vec{k}'}{(2\pi)^3}\, \tilde{V}(\vec{k}-\vec{k}')\, g(\vec{k}')$$

$$\int d\vec{r}\, e^{i\vec{r}(\vec{q}-\vec{k}+\vec{k}')} = (2\pi)^3\, \delta(\vec{q}-\vec{k}+\vec{k}')$$

$\Rightarrow$ Schrodinger eqn. in momentum space is

$$\frac{\hbar^2 k^2}{m} g(\vec{k}) + \int \frac{d^3\vec{k}'}{(2\pi)^3}\, \tilde{V}(\vec{k}-\vec{k}')\, g(\vec{k}') = \left(\varepsilon + \frac{\hbar^2 k_F^2}{m}\right) g(k)$$

Note $\tilde{V}(\vec{k}-\vec{k}') = \int d^3\vec{r}\; e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}}\; V(\vec{r})$.

In general, this interaction depends on directions of momenta ($\vec{k} = \hat{k}\,k$, $\vec{k}' = \hat{k}'\,k'$) on the Fermi surface

Can be expanded in spherical harmonics

$$\tilde{V}(\vec{k}-\vec{k}') = \sum_{\ell=0}^{\infty} V_\ell(k,k') \sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{k})\, Y^*_{\ell m}(\hat{k}')$$

spherical harmonics ~~[illegible]~~ orbital momentum ℓ, projection on z-axis m.

let $V_\ell(k,k') = \begin{cases} -V_\ell, & \text{for } \varepsilon_F < \frac{\hbar^2 k^2}{2m},\ \frac{\hbar^2 k'^2}{2m} < \varepsilon_F + \varepsilon_\ell \\ 0, & \text{outside this interval} \end{cases}$

(2) $$\left[\frac{\hbar^2}{m}\left(k^2 - k_F^2\right) - \varepsilon\right] g(k) = -\int \frac{d^3\vec{k}'}{(2\pi)^3} \sum_{\ell,m} (-V_\ell)\, Y_{\ell m}(\hat{k})\, Y^*_{\ell m}(\hat{k}') \cdot g(\vec{k}')$$

Expand $g(\vec{k}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} f_{\ell m}(k)\, Y_{\ell m}(\hat{k})$

and use orthonormality of $Y_{\ell m}$:

$$\int \frac{d\Omega'}{4\pi}\, Y^*_{\ell m}(\hat{k}')\, Y_{\ell' m'}(\hat{k}') = \delta_{\ell\ell'}\,\delta_{mm'}$$

Also use $\int \frac{d\vec{k}}{(2\pi)^3} \cdots = \underset{\substack{\| \\ \frac{m k_F}{2\pi^2\hbar^2} \\ \text{(density of states)}}}{N_0} \int d\xi \int \frac{d\Omega}{4\pi} \cdots$

$\xi \to \frac{\hbar^2 k^2}{2m} - \varepsilon_F$

$\frac{d\Omega}{4\pi} \to$ angular integration

RHS $$N_0 \int_0^{\varepsilon_\ell} d\xi' \int \frac{d\Omega'}{4\pi} \sum_{\ell,m} Y_{\ell m}(\hat{k}) Y^*_{\ell m}(\hat{k}') V_\ell \sum_{\ell',m'} f_{\ell' m'}(k') Y_{\ell' m'}(\hat{k}')$$

$$= N_0 \int d\xi' \sum_{\ell,m} V_\ell Y_{\ell m}(\hat{k}) f_{\ell m}(k')$$

$$\Rightarrow \left[\frac{\hbar^2}{m}(k^2-k_F^2) - \varepsilon\right] \sum_{\ell,m} f_{\ell m}(k) Y_{\ell m}(\hat{k}) =$$

$$= N_0 \int_0^{\varepsilon_\ell} d\xi' \sum_{\ell,m} V_\ell f_{\ell m}(k') Y_{\ell m}(\hat{k})$$

$$\Rightarrow \left[\underbrace{\frac{\hbar^2}{m}(k^2-k_F^2)}_{2\xi} - \varepsilon\right] f_{\ell m}(k) = N_0 V_\ell \int_0^{\varepsilon_\ell} d\xi' f_{\ell m}(k')$$

$$f_{\ell m}(k) = \frac{N_0 V_\ell \int_0^{\varepsilon_\ell} d\xi' f_{\ell m}(k')}{2\xi - \varepsilon}$$

$$\int_0^{\varepsilon_\ell} d\xi\, f_{\ell m}(k) = (N_0 V_\ell) \int_0^{\varepsilon_\ell} \frac{d\xi}{2\xi - \varepsilon} \int_0^{\varepsilon_\ell} d\xi' f_{\ell m}(k')$$

$$1 = N_0 V_\ell \int_0^{\varepsilon_\ell} \frac{d\xi}{2\xi - \varepsilon}$$

$$\int_0^{\varepsilon_\ell} \frac{d\xi}{2\xi-\varepsilon} = \frac{1}{2}\int_0^{2\varepsilon_\ell} \frac{dx}{x-\varepsilon} = \frac{1}{2}\ln\left(\frac{2\varepsilon_\ell-\varepsilon}{-\varepsilon}\right)$$

$$\Rightarrow 1 = N_0 V_\ell \, \frac{1}{2}\ln\left(\frac{\varepsilon_\ell - \frac{\varepsilon}{2}}{-\frac{\varepsilon}{2}}\right)$$

Exponentiate both sides

$$e^{+\frac{2}{N_0 V_\ell}} = \frac{\varepsilon_\ell - \frac{\varepsilon}{2}}{-\frac{\varepsilon}{2}}$$

$$-\frac{\varepsilon}{2}\, e^{\frac{2}{N_0 V_\ell}} = \varepsilon_\ell - \frac{\varepsilon}{2}$$

$$-\varepsilon_\ell = \frac{\varepsilon}{2}\left(e^{\frac{2}{N_0 V_\ell}} - 1\right)$$

Binding energy $\quad \varepsilon = \dfrac{-2\varepsilon_\ell}{e^{\frac{2}{N_0 V_\ell}}-1} \longrightarrow -2\varepsilon_\ell\, e^{-\frac{2}{N_0 V_\ell}}$

for $N_0 V_\ell \ll 1$

~~Binding~~ Energy gap $\quad \Delta_\ell = |\varepsilon| = 2\varepsilon_\ell\, e^{-2/N_0 V_\ell}$. (weak interaction)

$T_c \sim \Delta_\ell$. (crit. temp.)

Thus we find <u>bound</u> state $(\varepsilon < 0)$ of two electrons in channel with angular momentum ℓ.

Coherence length $\quad \xi \approx \dfrac{V_F}{|\varepsilon|} \approx \dfrac{V_F}{2\varepsilon_\ell}\, e^{2/N_0 V_\ell}$

$$\xi = \frac{V_F}{\Delta_\ell} \gg \text{inter-electron distance.}$$

Pair wave function:

$$g_\ell(\vec{k})\,\chi_{12}$$

χ_{12} ↳ spin part

$g_\ell(\vec{k})$ ↓

$f_{\ell m}(\vec{k})$ = orbital part = w.f. with angular momentum ℓ.

More precisely, $g_\ell(\vec{k}) = \sum_{m=-\ell}^{\ell} a_{\ell m}(k)\, Y_{\ell m}(\hat{k})$

$$g_\ell(-\vec{k}) = (-1)^\ell g_\ell(\vec{k}) \quad \text{(properties of } Y\text{)}$$

The pair wave function can be antisymmetric with respect to exchange of 2 particles:

$$g_\ell(\vec{k})\,\chi_{12} = -g_\ell(-\vec{k})\,\chi_{21}$$

$\Rightarrow$ for <u>even</u> $\ell = 0, 2, 4 \ldots$ (s, d, g) $\quad \chi_{12} = -\chi_{21}$ (spin w.f. anti symmetric)

for <u>odd</u> $\ell = 1, 3, \ldots$ (p, f) $\quad \chi_{12} = \chi_{21}$ (symmetric spin w.f.)

<u>S-wave</u> ($\ell = 0$), <u>d-wave</u> ($\ell = 2$) superconductors:

$\chi_{12} = -\chi_{21} \Rightarrow$ spin singlet state

$$\chi_{12} = |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \longrightarrow \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_y$$

$$\psi^{\ell}_{pair} = g(\vec{k})\, i\sigma_y = \sum_{m=-\ell}^{\ell} \underbrace{a_{\ell m}(k)}_{\text{plays role of the order parameter}} Y_{\ell m}(\hat{k})\, i\sigma_y$$

<u>P-wave</u> $(\ell=1)$ superconductor:

$\chi_{12} = \chi_{21} \Rightarrow$ spin triplet w.f.

$$\chi_{12} \Rightarrow \begin{cases} |\uparrow\uparrow\rangle \ (S_z = 1) \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ |\uparrow\downarrow + \downarrow\uparrow\rangle \ (S_z = 0) \rightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ |\downarrow\downarrow\rangle \ (S_z = -1) \rightarrow \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{cases}$$

$$\psi^{\ell=1}_{pair} = g_1(\vec{k})\,|\uparrow\uparrow\rangle + g_2(\vec{k})\,|\uparrow\downarrow + \downarrow\uparrow\rangle + g_3(\vec{k})\,|\downarrow\downarrow\rangle$$

$$\equiv \begin{pmatrix} g_1 & g_2 \\ g_2 & g_3 \end{pmatrix}$$

where $g_\alpha(\vec{k}) = \sum_{m=-\ell}^{\ell} a^{\alpha}_{\ell m}(k)\, Y_{\ell m}(\hat{k})$ are amplitudes of states with $S_z = 1, 0,$ and -1.

In the basis associated with matrices

$$i\vec{\sigma}\sigma_y = (i\,\sigma_x\sigma_y,\ i\sigma_y\sigma_y,\ i\sigma_z\sigma_y)$$

the same state reads

$$\psi^{\ell=1}_{pair} = i\,\vec{d}(\vec{k})\cdot\vec{\sigma}\,\sigma_y =$$

$$= (d_x\sigma_x + d_y\sigma_y + d_z\sigma_z)\,i\sigma_y = \begin{pmatrix} -d_x + i d_y & d_z \\ d_z & d_x + i d_y \end{pmatrix}$$